Qwen3.8-Flash-Next: which quant should YOU run?

10 local files · 3 quant/finetune labs · 20 effort arms · one measured RTX 5070 Ti (16 GB).

Ryzen 9800X3D · 64 GB RAM · Windows + WSL2 | Oct 7–12, 2026 local study | Engine column: Strata 0.1.40.3 / AgentionAI rc1 / rc2

Blue = ISTA-DASLab · Orange = UkisAI · Green = AgentionAI | GPT-6-luna MAX (OpenAI) · hosted reference only | PR #1850 · revised 2026-10-13

IN A HURRY? The study in six lines

- **Best measured stock all-rounder:** Q-Q2_0 (ISTA-DASLab) xhigh is #1 on the corrected eligible scoreboard (72). This is a seven-axis choice aid, not a universal intelligence score.
- **Pick the specialist when one axis matters.** Stock agent reliability: S-Q2_0 (UkisAI) low (78.3 floor). Fast coding: S-IQ3_XXS (UkisAI) low (11m56s). Those lower-composite picks do not contradict the all-rounder.
- **Coding ceilings are close; costs are not.** Highest complete reported score 96.3%: Q-IQ3_S (ISTA-DASLab) xhigh, Gyro-S (AgentionAI) low. Swift IQ3_S low reported the same percentage with only 163 scored tasks and is qualified, not tied on equal coverage.
- **Tool support is a serving-stack fact.** Gyro-M / rc2 joined the same pinned tool evaluator at 86.0 low and 87.8 xhigh. Gyro-S / rc1 remains n/a for tools. That is not an isolated model-only comparison.
- **More thinking has mixed returns.** Gyro-M xhigh solved one additional coding task after repair at 4.8x wall and 4.9x tokens. Its Hard Mode mean rose, but its floor fell from 78.3 to 71.7.
- **Fit and validity are part of the recommendation.** Gyro-M is experimental on this 16 GB card: 205 MiB minimum free VRAM, 78 C peak, invalid-slice status. No daily-driver endorsement. Partial scoring and test failures remain disclosed.

THE FOUR QUESTIONS — straight answers

Best overall on the measured stock rig?

Q-Q2_0 (ISTA-DASLab) xhigh · #1 overall. Same winner in the final board and first Pick Your File card.

Equal-weight composite; specialists can be better at the job you actually need.

Fastest complete coding loop?

S-IQ3_XXS (UkisAI) low · #9 overall. 11m56s at 95.1%.

This is elapsed coding-run time, not the fastest decoder.

Best stock agent reliability?

S-Q2_0 (UkisAI) low · #13 overall. Floor 78.3.

Gyro-M low ties the observed floor, but is a safety-capped experimental fork; no stock-composite rank.

Best local stock debugging / knowledge sample?

S-IQ3_S (UkisAI): hidden debugging 20/25, algebra sample 85%.

Low coding arm is partial/unranked; xhigh is #12. A local specialist, not the all-round crown.

PR #1850

THE CAST — ten files, one base model, three lineages

All local files share Qwen3.8-Flash-Next ancestry (Qwen / Alibaba). Q- labels: ISTA-DASLab quants; S- labels: UkisAI Swift derivatives; Gyro: AgentionAI APR. File sizes are decimal GB, not VRAM requirements. Results are integrated in the same tables; the OpenAI-hosted GPT-6-luna MAX reference appears alongside matched suites but is not one of these ten files, not a local run, and not a recommendation.

T01 · Model provenance and engine cohorts

File / lab	Quant	File GB	Engine	Status	Download
Q-IQ3_XXS ISTA-DASLab	IQ3_XXS	75.8	0.1.40.3	reference	HF repo
Q-Q2_0 ISTA-DASLab	Q2_0	66.4	0.1.40.3	reference	HF repo
Q-IQ3_S ISTA-DASLab	IQ3_S	83.6	0.1.40.3	reference	HF repo
Q-IQ2_XS ISTA-DASLab	IQ2_XS	67.2	0.1.40.3	reference	HF repo
S-IQ3_XXS UkisAI	IQ3_XXS	76.0	0.1.40.3	reference	HF repo
S-IQ2_XS UkisAI	IQ2_XS	68.2	0.1.40.3	reference	HF repo
S-IQ3_S UkisAI	IQ3_S	83.7	0.1.40.3	reference	HF repo
S-Q2_0 UkisAI	Q2_0	66.6	0.1.40.3	reference	HF repo
Gyro-S AgentionAI	TQ1_0	58.5	rc1	historical fork	HF repo
Gyro-M AgentionAI	TQ2_0	92.0	rc2	experimental	HF repo

T01 · 10 rows · end of table

The eight reference files use stock Strata 0.1.40.3; Gyro-S uses historical AgentionAI rc1, Gyro-M rc2. Shared evaluator and prompts do not remove the engine/configuration variable. Fork rows are inline observations, not ranked stock-engine controls.

THE SCORECARD — all models in one view

Tools: low-effort mean of five runs and empirical all-five floor. Coding: final percentage plus the actual solved/scored count. Asterisk = partial scoring. Knowledge/debugging are limited once-per-file tests. Stock rows sorted by floor; fork observations follow, with engine labels and no stock-composite rank.

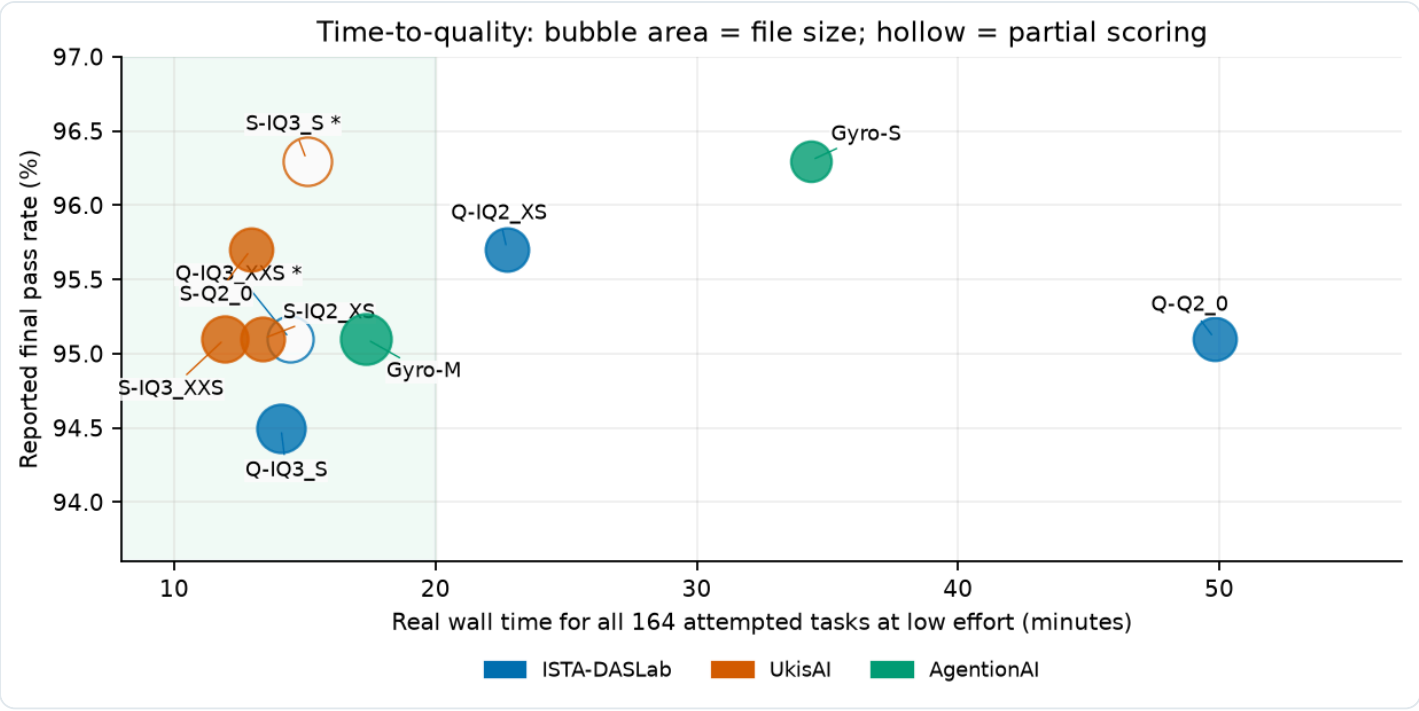
T02 · Integrated headline scorecard

File / lab	Engine	Tool mean	Pass^5	HE+ low	HE+ xhigh	Math	Algebra	Debug
S-Q2_0 UkisAI	0.1.40.3	85.0 +/- 1.0	78.3 1	95.7 (157/164)	93.3 (153/164)	90	72	18/25
Q-IQ3_XXS ISTA-DASLab	0.1.40.3	87.0 +/- 2.0	76.1 2	95.1 (155/163)*	95.7 (157/164)	99	79	16/25
S-IQ3_S UkisAI	0.1.40.3	88.2 +/- 1.5	76.1 2	96.3 (157/163)*	94.5 (155/164)	98	85	20/25
S-IQ3_XXS UkisAI	0.1.40.3	84.6 +/- 0.5	75.0 3	95.1 (156/164)	95.1 (156/164)	96	80	19/25
S-IQ2_XS UkisAI	0.1.40.3	87.2 +/- 1.3	75.0 3	95.1 (156/164)	94.5 (155/164)	97	83	18/25
Q-IQ3_S ISTA-DASLab	0.1.40.3	86.8 +/- 0.8	73.9	94.5 (155/164)	96.3 (158/164)	97	82	19/25
Q-IQ2_XS ISTA-DASLab	0.1.40.3	86.2 +/- 0.8	73.9	95.7 (157/164)	94.5 (155/164)	97	83	19/25
Q-Q2_0 ISTA-DASLab	0.1.40.3	84.8 +/- 0.4	70.7	95.1 (156/164)	95.1 (156/164)	97	78	19/25
Gyro-M AgentionAI	rc2	86.0 +/- 1.4	78.3	95.1 (156/164)	95.7 (157/164)	96	79	18/25
Gyro-S AgentionAI	rc1	n/a	n/a	96.3 (158/164)	94.4 (153/162)*	92	76	16/25

T02 · 10 rows · end of table

Medals show the top three distinct stock-reference score levels, ties share. Fork rows do not receive stock medals. n/a is an intentional unavailable measurement, not zero.

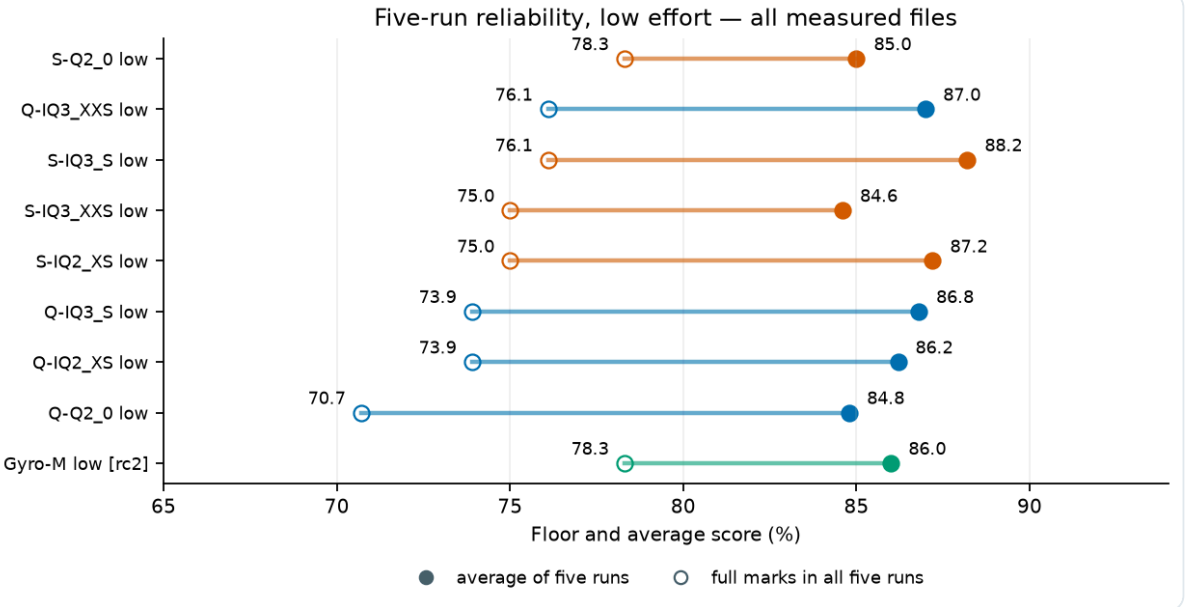
External anchor, not a pick: GPT-6-luna MAX (OpenAI) appears as a score-ordered reference row/bar in comparable suite tables and charts below, with no medal decoration. It did not run on the RTX 5070 Ti, is not a local file or quant, and is excluded from every local rank, recommendation and OVERALL composite.



Low-effort observed time-to-quality. Hollow points have partial scoring; green points use separate fork engines. No model-only causal speedup is claimed.

RELIABILITY & SAFETY — averages are not guarantees

Tool-eval-bench: 92 realistic scenarios x five identical-seed runs at each effort. Score 0 / 1 / 2; 2 = full marks. Pass@5: full marks at least once, the empirical capability ceiling. Pass^5: full marks in ALL five, the reliability floor. Partial credit every run never passes the floor. These summaries are not deployment guarantees. The table is ordered by Pass^5 floor (descending); mean breaks ties.



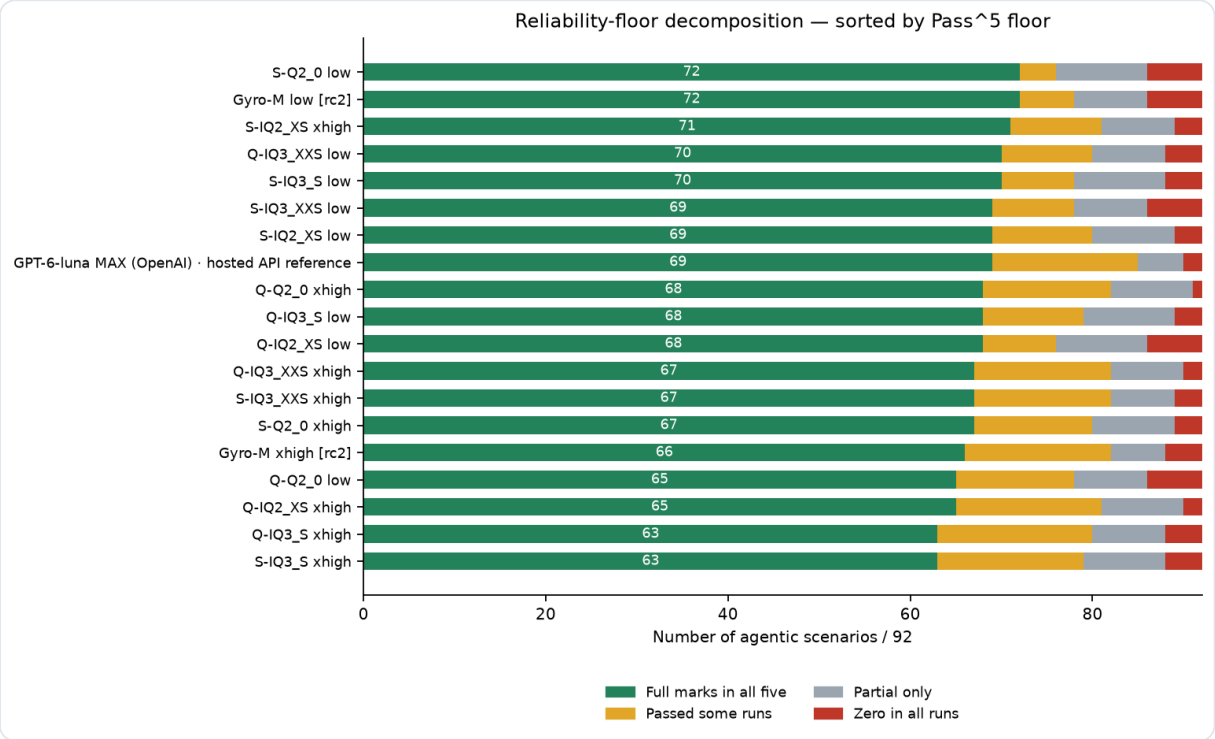
All measured low-effort files in the same reliability view. Engine labels preserve the stock/fork distinction. The line joins mean and floor; the table gap is Pass@5 minus Pass^5.

T03 · Integrated tool reliability — sorted by Pass^5 floor

Arm / lab	Mean +/- SD	Pass@5	Pass^5 floor	Ceiling-floor gap	Warnings (last trial)	Engine
Gyro-M AgentionAI low	86.0 +/- 1.4	84.8	78.3	6.5	6	rc2
S-Q2_0 UkisAI low	85.0 +/- 1.0	82.6	78.3 1	4.3	5	0.1.40.3
S-IQ2_XS UkisAI xhigh	88.2 +/- 1.5 3	88.0	77.2 2	10.8	2	0.1.40.3
S-IQ3_S UkisAI low	88.2 +/- 1.5 3	84.8	76.1 3	8.7	4	0.1.40.3
Q-IQ3_XXS ISTA-DASLab low	87.0 +/- 2.0	87.0	76.1 3	10.9	7	0.1.40.3
GPT-6-luna MAX OpenAI · hosted API reference	90.0 +/- 1.9	92.4	75.0	17.4	0	OpenAI Codex Responses API · max
S-IQ2_XS UkisAI low	87.2 +/- 1.3	87.0	75.0	12.0	4	0.1.40.3
S-IQ3_XXS UkisAI low	84.6 +/- 0.5	84.8	75.0	9.8	4	0.1.40.3
Q-Q2_0 ISTA-DASLab xhigh	89.4 +/- 0.5 1	89.1	73.9	15.2	2	0.1.40.3
Q-IQ3_S ISTA-DASLab low	86.8 +/- 0.8	85.9	73.9	12.0	3	0.1.40.3
Q-IQ2_XS ISTA-DASLab low	86.2 +/- 0.8	82.6	73.9	8.7	5	0.1.40.3
Q-IQ3_XXS ISTA-DASLab xhigh	88.6 +/- 1.1 2	89.1	72.8	16.3	3	0.1.40.3
S-Q2_0 UkisAI xhigh	86.8 +/- 2.2	87.0	72.8	14.2	2	0.1.40.3
S-IQ3_XXS UkisAI xhigh	86.6 +/- 1.5	89.1	72.8	16.3	4	0.1.40.3
Gyro-M AgentionAI xhigh	87.8 +/- 1.1	89.1	71.7	17.4	2	rc2
Q-IQ2_XS ISTA-DASLab xhigh	88.6 +/- 1.8 2	88.0	70.7	17.3	1	0.1.40.3
Q-Q2_0 ISTA-DASLab low	84.8 +/- 0.4	84.8	70.7	14.1	7	0.1.40.3
Q-IQ3_S ISTA-DASLab xhigh	86.8 +/- 1.3	87.0	68.5	18.5	4	0.1.40.3
S-IQ3_S UkisAI xhigh	85.6 +/- 0.5	85.9	68.5	17.4	5	0.1.40.3

T03 · 19 rows · end of table

The hosted row is MAX effort across 92 scenarios x five trials. Confidence intervals and evaluator rating are omitted because the hosted artifact does not publish comparable values; no proxy is inferred. Warning counts describe the final primary trial, not a five-run union. GPT-6-luna MAX is an external reference, excluded from the local recommendation cohort. 8/16 stock arms are capped on the stored last-trial rating. Gyro-M adds one capped and one Good rating. Not capped is NOT the same as no warnings or production safety. Warning lists/rating describe the final primary trial, not a five-run union.



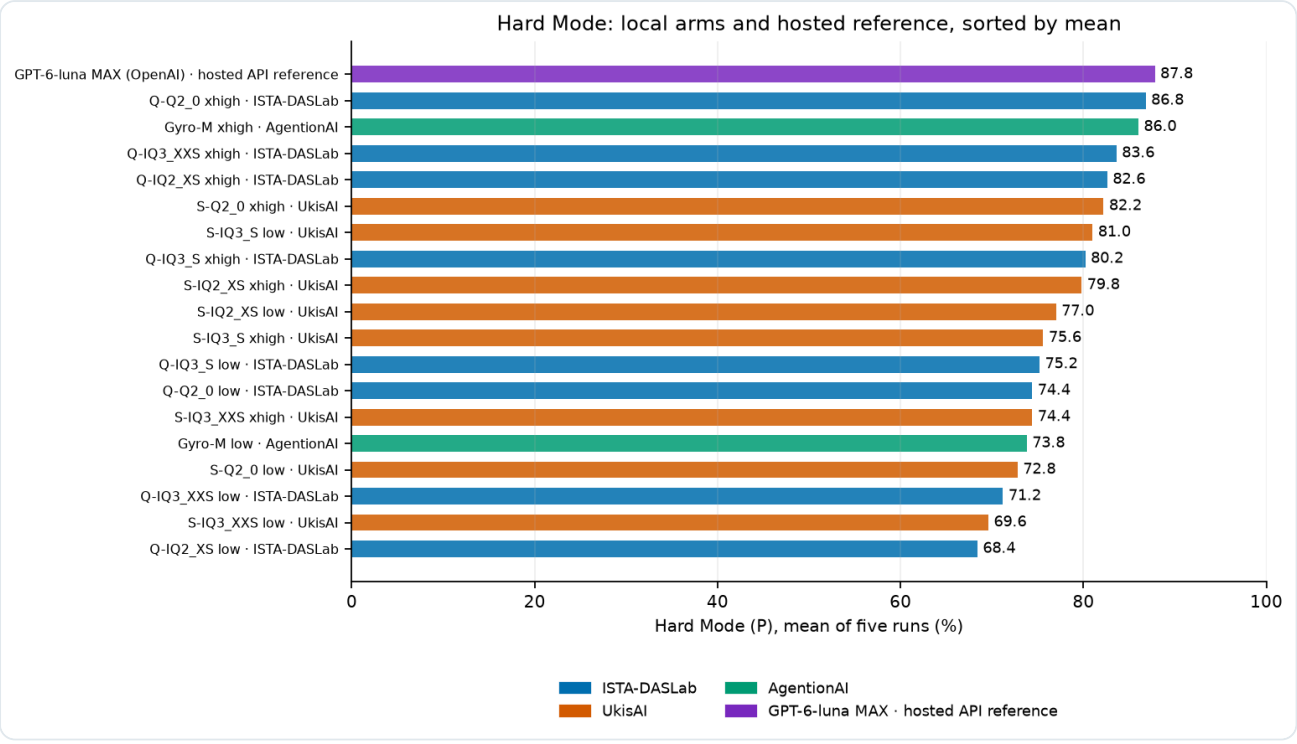
The all-five floor is the green count / 92. Rows are ordered by that floor; GPT-6-luna MAX is interleaved at its score and remains outside the local scorecard and composite.

> Gyro-M illustrates the trade-off: always-full scenarios 72 low / 66 xhigh; flaky 6 / 16. Its mean confidence intervals overlap, so the observed mean gain is modest. No causal mechanism is inferred from the aggregate alone.

Tool-eval-bench

AGENTIC TOOL USE — hard chains reward extra thought

Hard Mode (P) stresses long chains and recovery. Every category figure uses the five-run mean, not the last-trial score. Effort comparisons stay within a measured file/build. The table is ordered by Hard Mode mean (descending); GPT-6-luna MAX is placed by its score but remains an external hosted reference.



Every local effort arm and GPT-6-luna MAX are shown as score-ordered bars. The purple row is an external hosted reference, not a local model.

T04 · Multi-step tool-chain results — sorted by Hard Mode mean

Arm / lab	Hard Mode mean	SD across five	Turn budget	Engine
GPT-6-luna MAX OpenAI · hosted API reference	87.8	7.0	8	OpenAI Codex Responses API · max
Q-Q2_0 ISTA-DASLab xhigh	86.8 1	4.8	8	0.1.40.3
Gyro-M AgentionAI xhigh	86.0	4.7	8	rc2
Q-IQ3_XXS ISTA-DASLab xhigh	83.6 2	2.6	8	0.1.40.3
Q-IQ2_XS ISTA-DASLab xhigh	82.6 3	4.9	8	0.1.40.3
S-Q2_0 UkisAI xhigh	82.2	4.3	8	0.1.40.3
S-IQ3_S UkisAI low	81.0	3.1	8	0.1.40.3
Q-IQ3_S ISTA-DASLab xhigh	80.2	3.9	8	0.1.40.3
S-IQ2_XS UkisAI xhigh	79.8	3.0	8	0.1.40.3
S-IQ2_XS UkisAI low	77.0	5.1	8	0.1.40.3
S-IQ3_S UkisAI xhigh	75.6	3.0	8	0.1.40.3
Q-IQ3_S ISTA-DASLab low	75.2	1.1	8	0.1.40.3
Q-Q2_0 ISTA-DASLab low	74.4	2.2	8	0.1.40.3
S-IQ3_XXS UkisAI xhigh	74.4	3.3	8	0.1.40.3
Gyro-M AgentionAI low	73.8	4.4	8	rc2
S-Q2_0 UkisAI low	72.8	3.3	8	0.1.40.3
Q-IQ3_XXS ISTA-DASLab low	71.2	7.2	8	0.1.40.3
S-IQ3_XXS UkisAI low	69.6	3.6	8	0.1.40.3
Q-IQ2_XS ISTA-DASLab low	68.4	2.3	8	0.1.40.3

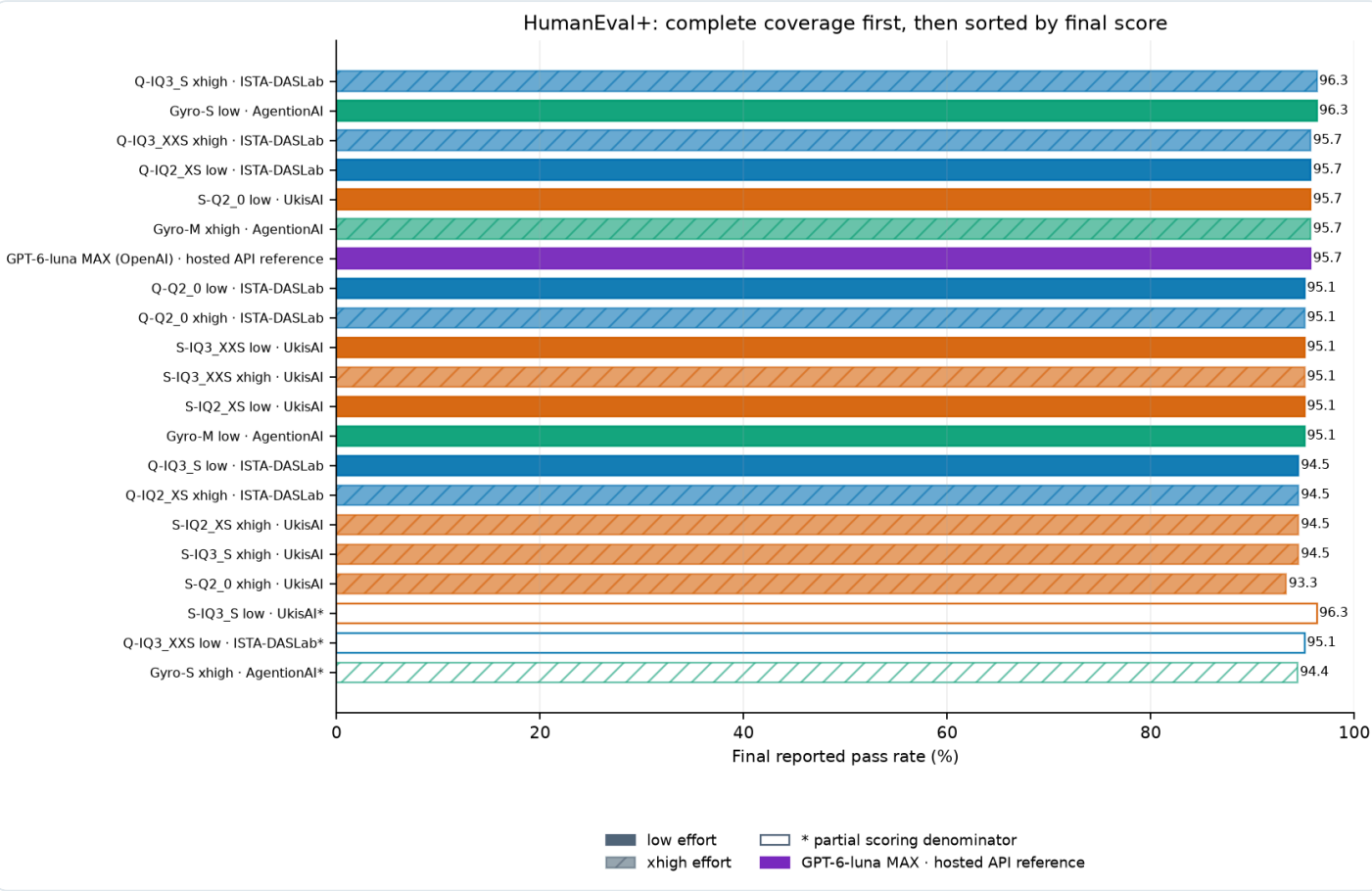
T04 · 19 rows · end of table

- Gyro-M Hard Mode rises from 73.8% to 86.0%, while its all-five floor falls. Optimizing only one of those axes is not a reliable workload decision.
- Warnings include premature notifications, missed dependency ordering and claims that an invoice is paid while funds remain held. They are scripted failure traps, not permissions.

- rc2 live forced tool calls passed streaming and non-streaming with Gyro-M; Gyro-S rc1 remains unavailable for tool-eval. This does not establish Gyro-S behavior on rc2.

CODE — every denominator visible

164 function-writing tasks via BenchKit / EvalPlus, greedy generation, one verifier-guided repair. First = one-shot score; final includes the replacement. Scored timeout/length/blank outputs remain failures. Excluded harness errors can reduce the denominator; those rows retain observed percentages but are qualified and excluded from full-coverage/composite claims. The table orders complete-coverage runs by final score, followed by partial-coverage runs.



Complete-coverage rows are score-sorted first; partial-coverage rows follow. GPT-6-luna MAX is shown at its final-score position as a hosted reference. * marks partial scoring coverage.

T05 · Integrated coding results — sorted by final score

Arm / lab	First %	Final %	Solved / scored	Repairs	Wall	tok/s	Blank finals	Infra excluded	Engine
Q-IQ3_S ISTA-DASLab xhigh	93.3	96.3 1	158/164	5/11	67m19s	86.7	2	0	0.1.40.3
Gyro-S AgentionAI low	92.1	96.3	158/164	7/13	34m21s	71.7	2	0	rc1
Q-IQ3_XXS ISTA-DASLab xhigh	92.7	95.7 2	157/164	5/11	76m17s	102.6	3	0	0.1.40.3
Q-IQ2_XS ISTA-DASLab low	90.9	95.7 2	157/164	8/14	22m44s	124.5	2	0	0.1.40.3
S-Q2_0 UkisAI low	90.2	95.7 2	157/164	9/16	12m56s 2	144.8	0	0	0.1.40.3
Gyro-M AgentionAI xhigh	92.1	95.7	157/164	6/13	82m51s	77.6	2	0	rc2
GPT-6-luna MAX OpenAI · hosted API reference	93.3	95.7	157/164	4/11	33m08s	42.0†	0	0	OpenAI Codex Responses API · max
Q-Q2_0 ISTA-DASLab low	92.7	95.1 3	156/164	4/10	49m50s	123.9	2	0	0.1.40.3
Q-Q2_0 ISTA-DASLab xhigh	88.4	95.1 3	156/164	11/17	62m18s	124.0	3	0	0.1.40.3
S-IQ3_XXS UkisAI low	92.7	95.1 3	156/164	4/12	11m56s 1	123.7	0	0	0.1.40.3
S-IQ3_XXS UkisAI xhigh	92.1	95.1 3	156/164	5/10	47m11s	104.5	5	0	0.1.40.3
S-IQ2_XS UkisAI low	89.6	95.1 3	156/164	9/16	13m23s 3	148.8	1	0	0.1.40.3
Gyro-M AgentionAI low	93.3	95.1	156/164	3/11	17m20s	88.9	0	0	rc2
Q-IQ3_S ISTA-DASLab low	92.7	94.5	155/164	3/12	14m05s	100.4	0	0	0.1.40.3
Q-IQ2_XS ISTA-DASLab xhigh	89.6	94.5	155/164	8/16	75m40s	122.0	6	0	0.1.40.3
S-IQ2_XS UkisAI xhigh	91.5	94.5	155/164	5/11	63m04s	124.0	5	0	0.1.40.3
S-IQ3_S UkisAI xhigh	92.7	94.5	155/164	3/11	43m32s	89.5	3	0	0.1.40.3
S-Q2_0 UkisAI xhigh	89.6	93.3	153/164	6/14	56m20s	121.8	6	0	0.1.40.3
S-IQ3_S UkisAI low	94.5	96.3*	157/163	3/9	15m05s	101.5	0	1	0.1.40.3
Q-IQ3_XXS ISTA-DASLab low	92.6	95.1*	155/163	4/13	14m26s	121.5	0	1	0.1.40.3
Gyro-S AgentionAI xhigh	90.1	94.4*	153/162	7/16	329m15s	63.1	3	2	rc1

T05 · 21 rows · end of table

† GPT-6-luna MAX 42.0 tok/s is provider-reported completion usage and includes hidden reasoning tokens; it is not visible-text throughput, local RTX 5070 Ti decode speed, or a matched speed result. No hosted performance rates are published.

Retrospective coverage correction: Q-IQ3_XXS low 155/163; Swift IQ3_S low 157/163; Gyro-S xhigh 153/162 after two HTTP 502 harness errors. Their observed percentages are preserved, not silently rescored. Gyro-M xhigh blank finals on HumanEval/39 and HumanEval/99 ended on length and remain scored failures; 157/164 is not a clean-response certificate.

- Gyro-M xhigh costs 4.8x low wall and 4.9x tokens for one additional final pass. Hard Mode benefits do not imply a coding-efficiency benefit.
- Gyro-M low vs historical Gyro-S low has 1.2 points more first-attempt score, 1.2 points less final score. Engines/configurations differ; this is not isolated quant improvement.

EvalPlus BenchKit

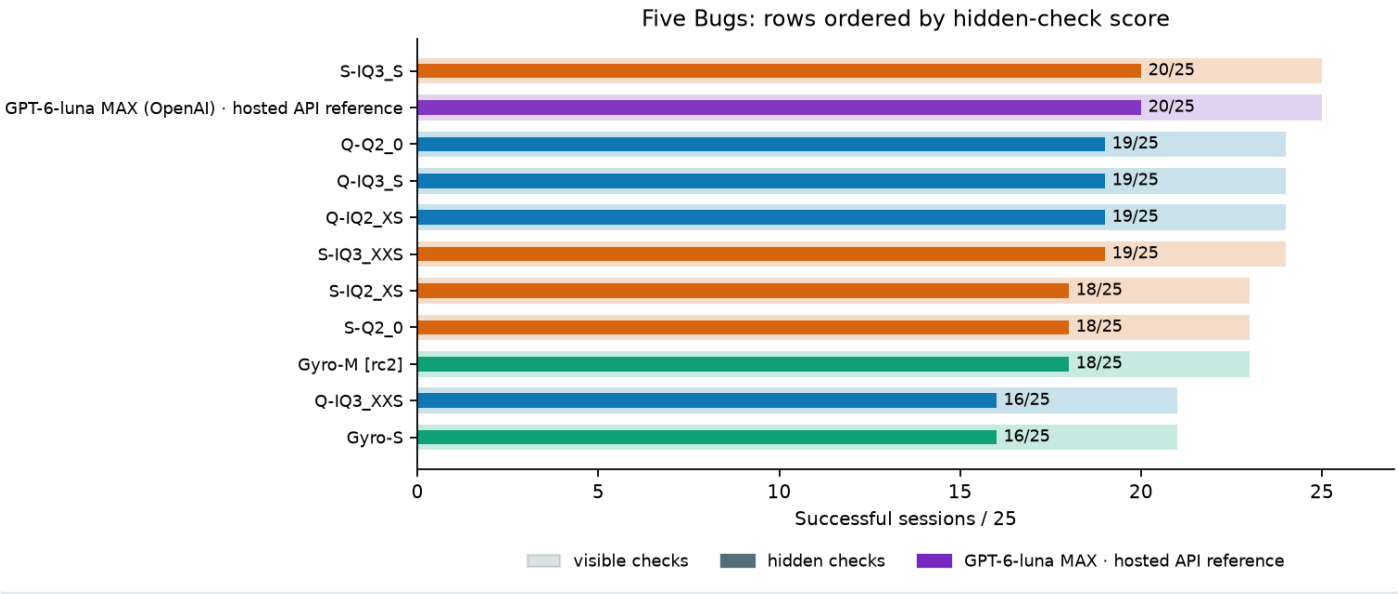
KNOWLEDGE & DEBUGGING — the test we actually ran

GSM8K: 100 seeded-shuffled questions, 8-shot. MMLU: first 100, 5-shot, entirely abstract algebra — NOT a 57-subject exam. IFEval: first 50 prompts / 76 constraints. Five Bugs: five buggy Python files x five sessions, visible plus local hidden checks. All are limited samples; do not turn them into a whole-model intelligence ranking. Rows are ordered by hidden debugging passes (descending); the hosted reference is interleaved by score, not included in the local composite.

T06 · Integrated sampled suites — sorted by hidden debugging passes

File / lab	GSM8K /100	Algebra /100	IFEval prompt / instruction %	Bugs visible	Bugs hidden	Needle % @ tokens	Engine
S-IQ3_S UkisAI	98	85	84.0 / 89.5	25/25	20/25 1	100 @ 260,096	0.1.40.3
GPT-6-luna MAX OpenAI · hosted API reference	97	93	86.0 / 90.8	25/25	20/25	100 @ 260,096	OpenAI Codex Responses API · max
Q-Q2_0 ISTA-DASLab	97	78	84.0 / 89.5	24/25	19/25 2	100 @ 260,096	0.1.40.3
Q-IQ3_S ISTA-DASLab	97	82	86.0 / 90.8	24/25	19/25 2	100 @ 260,096	0.1.40.3
Q-IQ2_XS ISTA-DASLab	97	83	84.0 / 89.5	24/25	19/25 2	100 @ 260,096	0.1.40.3
S-IQ3_XXS UkisAI	96	80	80.0 / 86.8	24/25	19/25 2	100 @ 260,096	0.1.40.3
S-IQ2_XS UkisAI	97	83	82.0 / 88.2	23/25	18/25 3	100 @ 260,096	0.1.40.3
S-Q2_0 UkisAI	90	72	80.0 / 86.8	23/25	18/25 3	100 @ 260,096	0.1.40.3
Gyro-M AgentionAI	96	79	78.0 / 82.9	23/25	18/25	100 @ 129,024	rc2
Q-IQ3_XXS ISTA-DASLab	99	79	80.0 / 85.5	21/25	16/25	100 @ 260,096	0.1.40.3
Gyro-S AgentionAI	92	76	74.0 / 82.9	21/25	16/25	100 @ 129,024	rc1

T06 · 11 rows · end of table



Ten local files plus GPT-6-luna MAX; the purple bars are the separate OpenAI-hosted API reference, not a local quant.

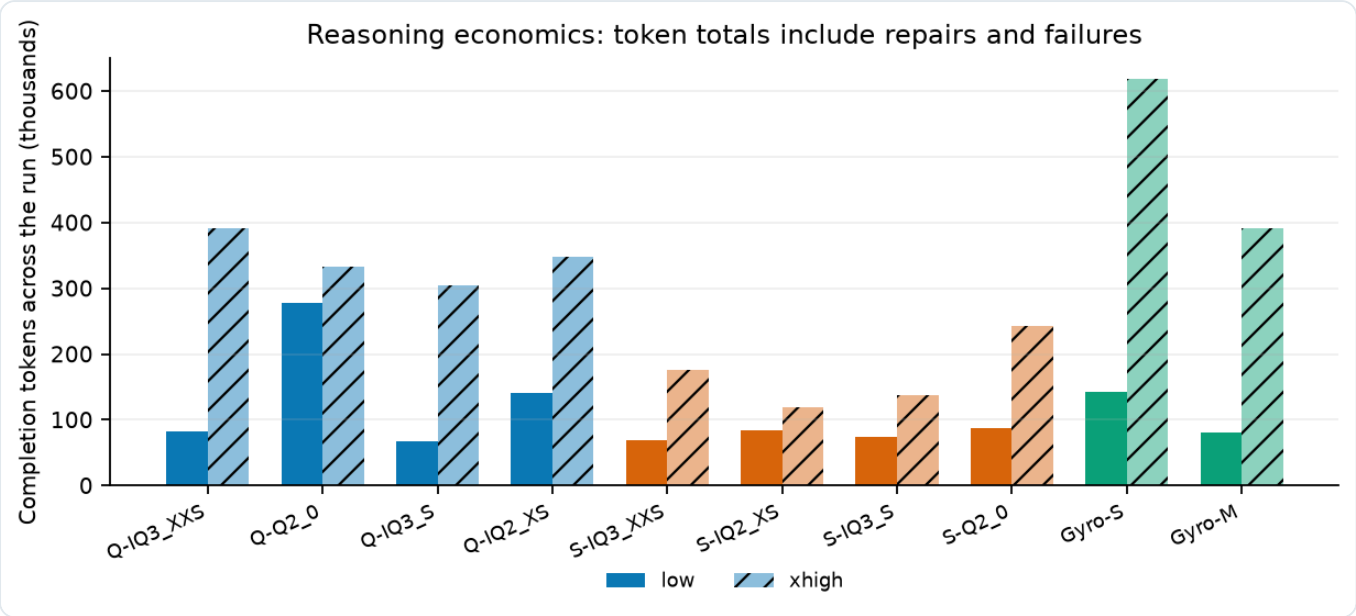
Within the local stock cohort, S-IQ3_S (UkisAI) remains the debugging specialist at 20/25 hidden sessions and leads this algebra sample at 85%. GPT-6-luna MAX posts 20/25 hidden checks as an external reference, not a local quant candidate.

Needle: 25 cells (five lengths x five insertion depths). Effective retrieved lengths are 260,096 for stock and 129,024 for both fork rows. Gyro-M configured context 131,072; no 260K claim for it.

Five Bugs upstream | TEB knowledge / needle plugins

REASONING ECONOMICS — what you wait for

Total output tokens include reasoning, repairs and failures. Wall includes request/harness overhead. Failure time share and passing-task cost come from task-level response times, not wall subtraction. Different engine/protocol rows remain descriptive.



Completion-token totals for all attempted tasks, all files, both efforts. Partial-scoring arms retain their coverage qualifier.

T07 · Integrated time-to-result economics

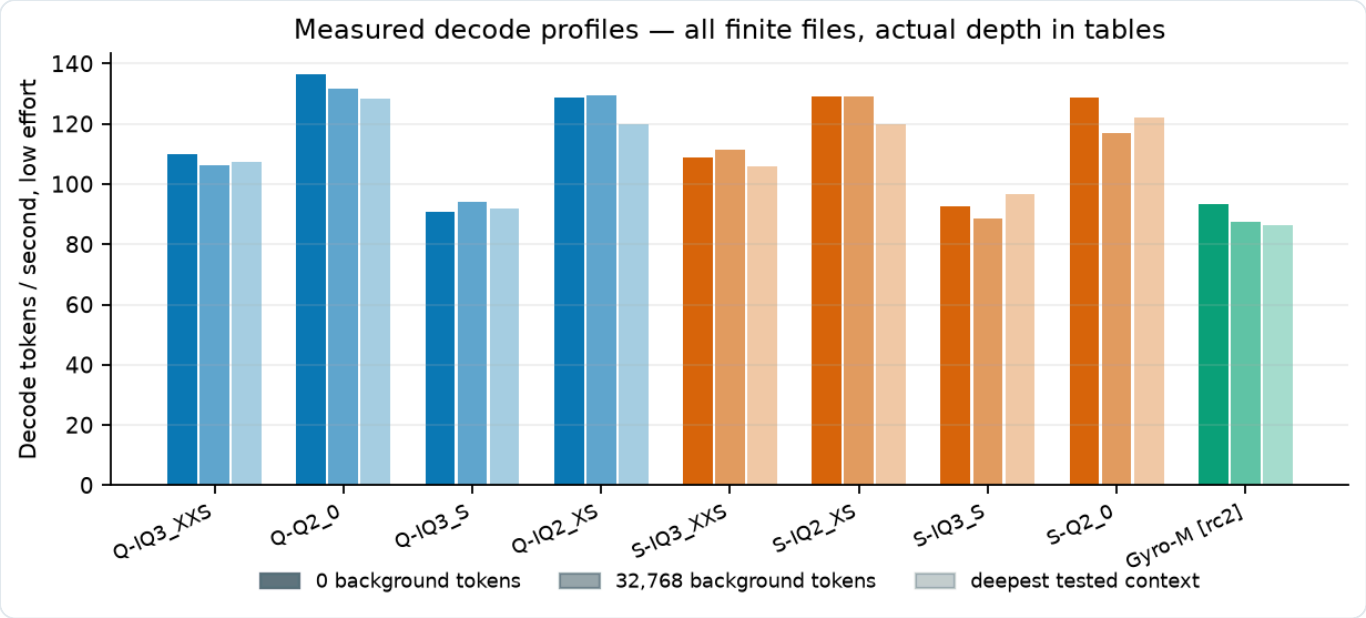
Arm / lab	Output tokens	Passing tokens	Passing seconds	Failed response time %	Wall	Engine
S-IQ3_XXS UkisAI low	68,616	49,192	486	25.1%	11m56s	0.1.40.3
S-Q2_0 UkisAI low	87,634	64,502	542	23.4%	12m56s	0.1.40.3
S-IQ2_XS UkisAI low	83,746	56,476	468	36.3%	13m23s	0.1.40.3
Q-IQ3_S ISTA-DASLab low	66,966	49,239	594	23.9%	14m05s	0.1.40.3
Q-IQ3_XXS ISTA-DASLab low	82,843	59,863	602	24.7%	14m26s	0.1.40.3
S-IQ3_S UkisAI low	74,417	49,341	598	29.0%	15m05s	0.1.40.3
Q-IQ2_XS ISTA-DASLab low	140,583	97,239	883	32.1%	22m44s	0.1.40.3
S-IQ3_S UkisAI xhigh	137,839	81,261	1,017	60.2%	43m32s	0.1.40.3
S-IQ3_XXS UkisAI xhigh	175,860	133,095	1,363	50.7%	47m11s	0.1.40.3
Q-Q2_0 ISTA-DASLab low	277,084	174,148	1,498	48.8%	49m50s	0.1.40.3
S-Q2_0 UkisAI xhigh	243,330	151,062	1,329	60.0%	56m20s	0.1.40.3
Q-Q2_0 ISTA-DASLab xhigh	332,296	238,341	2,020	45.1%	62m18s	0.1.40.3
S-IQ2_XS UkisAI xhigh	118,588	105,555	2,472	33.6%	63m04s	0.1.40.3
Q-IQ3_S ISTA-DASLab xhigh	304,962	200,835	2,461	38.1%	67m19s	0.1.40.3
Q-IQ2_XS ISTA-DASLab xhigh	347,704	257,698	2,214	50.5%	75m40s	0.1.40.3
Q-IQ3_XXS ISTA-DASLab xhigh	390,930	230,123	2,328	48.5%	76m17s	0.1.40.3
Gyro-M AgentionAI low	80,523	60,016	752	23.2%	17m20s	rc2
Gyro-S AgentionAI low	143,288	69,471	1,316	34.1%	34m21s	rc1
Gyro-M AgentionAI xhigh	391,467	180,234	2,334	52.5%	82m51s	rc2
Gyro-S AgentionAI xhigh	618,982	375,211	6,148	60.1%	329m15s	rc1

T07 · 20 rows · end of table

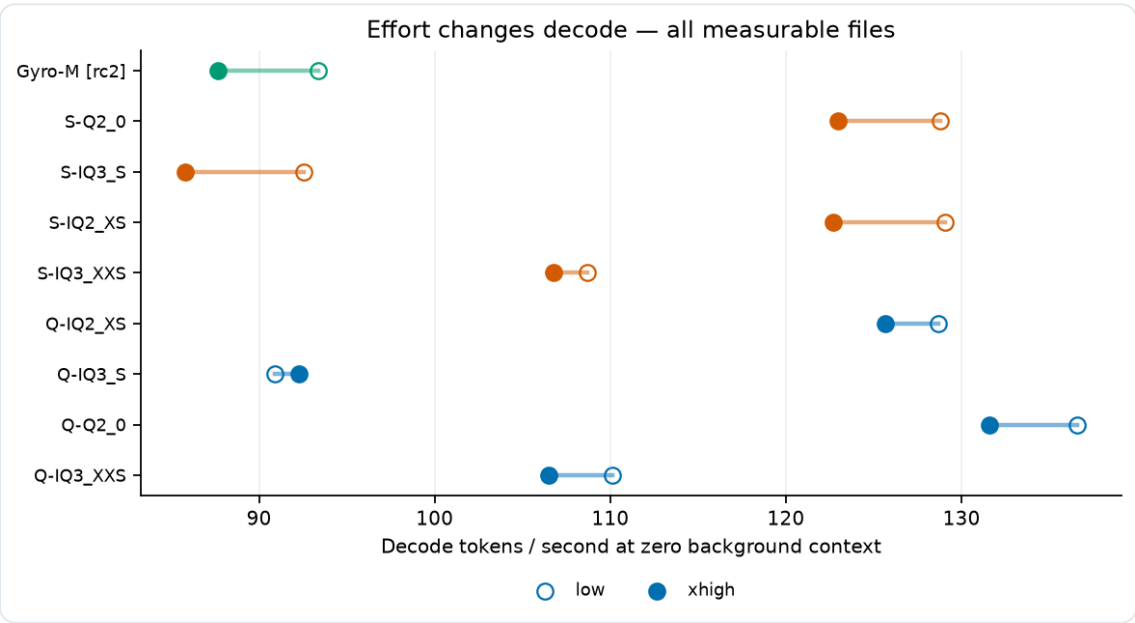
Pick effort for the task: more deliberation can improve long tool chains while making code loops slower. The overall ranking averages capability axes, not latency or safety; the specialist cards below say exactly which axis overrides that ranking.

SPEED — real decode; corrected first-token timing

llama-benchy 0.4.0: prompt 2,048, generation 256, concurrency 1, three repeats/depth/effort. Raw pp_throughput in the old PDF was inflated by early SSE role chunks and is retired. Here prompt-to-first-token rate = (prompt + background context) / e2e TTFT, including network/queue/reasoning. It is NOT an engine-kernel prefill benchmark. Largest depths differ: stock 131,072; rc2 122,880.



All finite low-effort decode profiles. Lightest bar = deepest measured point; row tables state the exact token depths. Gyro-S malformed profiles remain n/a.



Decode at zero background context for both efforts, all measurable files. A fork-engine observation is not a same-engine model A/B.

T08 · Integrated low-effort performance profiles

File / lab	Engine	Context tokens: 0 / mid / deep	Decode tok/s: 0 / mid / deep	First-token rate: 0 / mid / deep	Deep TTFT s
Q-Q2_0 ISTA-DASLab	0.1.40.3	0 / 32,768 / 131,072	136.6 / 131.8 / 128.4	2,309 / 3,522 / 3,667	36.30
S-IQ2_XS UkisAI	0.1.40.3	0 / 32,768 / 131,072	129.1 / 129.1 / 119.9	1,970 / 3,283 / 3,334	39.93
S-Q2_0 UkisAI	0.1.40.3	0 / 32,768 / 131,072	128.8 / 116.9 / 121.9	1,892 / 3,229 / 3,329	39.99
Q-IQ2_XS ISTA-DASLab	0.1.40.3	0 / 32,768 / 131,072	128.7 / 129.4 / 120.0	2,011 / 3,195 / 3,280	40.59
Q-IQ3_XXS ISTA-DASLab	0.1.40.3	0 / 32,768 / 131,072	110.1 / 106.2 / 107.4	2,174 / 3,284 / 3,421	38.91
S-IQ3_XXS UkisAI	0.1.40.3	0 / 32,768 / 131,072	108.7 / 111.3 / 106.0	2,025 / 3,329 / 3,412	39.01
S-IQ3_S UkisAI	0.1.40.3	0 / 32,768 / 131,072	92.6 / 88.5 / 96.5	1,594 / 3,206 / 3,374	39.45
Q-IQ3_S ISTA-DASLab	0.1.40.3	0 / 32,768 / 131,072	90.9 / 94.0 / 91.7	1,641 / 3,164 / 3,041	43.77
Gyro-M AgentionAI	rc2	0 / 32,768 / 122,880	93.4 / 87.5 / 86.5	1,239 / 2,144 / 2,821	44.28
Gyro-S AgentionAI	rc1	n/a	n/a	n/a	n/a

T08 · 10 rows · end of table

T09 · Integrated xhigh-effort performance profiles

File / lab	Engine	Context tokens: 0 / mid / deep	Decode tok/s: 0 / mid / deep	First-token rate: 0 / mid / deep	Deep TTFT s
Q-Q2_0 ISTA-DASLab	0.1.40.3	0 / 32,768 / 131,072	131.6 / 130.5 / 117.2	2,676 / 3,836 / 3,663	36.35
Q-IQ2_XS ISTA-DASLab	0.1.40.3	0 / 32,768 / 131,072	125.7 / 118.9 / 117.5	2,135 / 3,341 / 3,339	39.87
S-Q2_0 UkisAI	0.1.40.3	0 / 32,768 / 131,072	123.0 / 121.9 / 128.9	2,219 / 3,167 / 3,332	39.96
S-IQ2_XS UkisAI	0.1.40.3	0 / 32,768 / 131,072	122.7 / 123.9 / 113.8	2,209 / 3,286 / 3,334	39.93
S-IQ3_XXS UkisAI	0.1.40.3	0 / 32,768 / 131,072	106.8 / 101.1 / 99.4	2,420 / 3,339 / 3,449	38.59
Q-IQ3_XXS ISTA-DASLab	0.1.40.3	0 / 32,768 / 131,072	106.5 / 101.7 / 95.3	2,406 / 3,361 / 3,409	39.05
Q-IQ3_S ISTA-DASLab	0.1.40.3	0 / 32,768 / 131,072	92.2 / 91.2 / 86.6	1,945 / 3,216 / 3,382	39.36
S-IQ3_S UkisAI	0.1.40.3	0 / 32,768 / 131,072	85.8 / 85.5 / 85.8	1,933 / 3,269 / 3,388	39.29
Gyro-M AgentionAI	rc2	0 / 32,768 / 122,880	87.7 / 80.5 / 80.7	1,754 / 2,690 / 2,778	44.96
Gyro-S AgentionAI	rc1	n/a	n/a	n/a	n/a

T09 · 10 rows · end of table

> Gyro-S synthetic streaming produced million-token/s decode artifacts and null depth cells; those profiles are excluded. GPT-6-luna MAX matched performance is unavailable for matched decode comparison: requested [3, 3, 3] repeats at contexts [0, 32768, 122880], but only [1, 2, 3] valid visible-text rate samples were retained; all hosted rates are withheld and excluded from local decode plots. The provider-reported HE+ token rate above is not a substitute.

THE FINAL SCOREBOARD — one message, with explicit scope

OVERALL averages seven min-max-normalized axes within the 14 eligible stock-engine arms: coding final, agent floor, Hard Mode mean, algebra sample, math sample, hidden debugging, zero-context decode. Full scoring only. Forks and partial arms appear inline but are unranked; the GPT-6-luna MAX hosted reference is shown in suite tables only and is excluded from this local composite. This crowns the best measured STOCK ALL-ROUNDER, not every workload winner; safety and latency are not averaged into that score.

Decision board: stock-engine composite only; forks and partial arms unranked

	Rank / OVERALL	Coding HE+	Agents Pass^5	Hard Mode	MMLU sample	GSM8K sample	Debug /25	Decode tok/s	Rating gate
Q-Q2_0 xhigh	1. 72	95.1	73.9	86.8	78	97	19	132	not capped
S-IQ2_XS xhigh	2. 68	94.5	77.2	79.8	83	97	18	123	not capped
S-IQ2_XS low	3. 67	95.1	75.0	77.0	83	97	18	129	capped
Q-IQ2_XS low	4. 65	95.7	73.9	68.4	83	97	19	129	capped
Q-IQ2_XS xhigh	5. 65	94.5	70.7	82.6	83	97	19	126	not capped
Q-Q2_0 low	6. 59	95.1	70.7	74.4	78	97	19	137	capped
Q-IQ3_S xhigh	7. 58	96.3	68.5	80.2	82	97	19	92	capped
Q-IQ3_XXS xhigh	8. 57	95.7	72.8	83.6	79	99	16	106	capped
S-IQ3_XXS low	9. 54	95.1	75.0	69.6	80	96	19	109	not capped
S-IQ3_XXS xhigh	10. 54	95.1	72.8	74.4	80	96	19	107	capped
Q-IQ3_S low	11. 53	94.5	73.9	75.2	82	97	19	91	not capped
S-IQ3_S xhigh	12. 53	94.5	68.5	75.6	85	98	20	86	capped
S-Q2_0 low	13. 48	95.7	78.3	72.8	72	90	18	129	not capped
S-Q2_0 xhigh	14. 35	93.3	72.8	82.2	72	90	18	123	not capped
Q-IQ3_XXS low	unranked	95.1*	76.1	71.2	79	99	16	110	capped
S-IQ3_S low	unranked	96.3*	76.1	81.0	85	98	20	93	not capped
Gyro-S low	unranked	96.3	n/a	n/a	76	92	16	n/a	n/a
Gyro-S xhigh	unranked	94.4*	n/a	n/a	76	92	16	n/a	n/a
Gyro-M low	unranked	95.1	78.3	73.8	79	96	18	93	capped
Gyro-M xhigh	unranked	95.7	71.7	86.0	79	96	18	88	not capped

Same ranks and scope used by every recommendation below. Asterisk = partial coding denominator. Not capped is a final-trial rating status, not a safety guarantee.

All-rounder: Q-Q2_0 (ISTA-DASLab) xhigh · #1 overall (72). **Specialists:** S-Q2_0 (UkisAI) low · #13 owns the stock floor; S-IQ3_XXS (UkisAI) low · #9 owns the fastest complete coding wall. Pick Your File translates THIS board; it does not introduce a second overall winner.

T10 · One recommendation registry — all-rounder versus specialists

Decision	File / effort	OVERALL rank	Why this agrees with the board
Best stock all-rounder	**Q-Q2_0** ISTA-DASLab xhigh	#1	Balance the seven measured capability axes
Reliability specialist	**S-Q2_0** UkisAI low	#13	Prioritize the empirical tool floor
Coding-latency specialist	**S-IQ3_XXS** UkisAI low	#9	Prioritize complete-run wall time, outside OVERALL
Coding-ceiling specialist	**Q-IQ3_S** ISTA-DASLab xhigh	#7	Prioritize the complete coding score
Knowledge / debugging specialist	**S-IQ3_S** UkisAI xhigh	#12	Prioritize independent sampled tests; low is partial/unranked
Experimental observation only	**Gyro-M** AgentionAI low	unranked	Different engine and narrow 16 GB margin; no production pick

T10 · 6 rows · end of table

PICK YOUR FILE — the scoreboard translated into jobs

One decision rule: use the #1 stock all-rounder if you balance the seven measured axes. Override that composite only when your workload has a named priority — reliability, coding latency, coding ceiling, or debugging. Lower-ranked specialists are deliberate exceptions, not alternative overall crowns. Unranked experimental/partial rows are not eligible production recommendations.

Balanced workload · best stock all-rounder

Q-Q2_0 (ISTA-DASLab) xhigh · #1 overall

Score 72 across the seven eligible stock axes. Also leads stock tool mean at 89.4.

Composite is equal-weighted, on one rig and limited knowledge samples; not a universal quality/safety claim.

Reliability-first agent · specialist override

S-Q2_0 (UkisAI) low · #13 overall

Stock floor 78.3 is highest. Choose this lower-ranked arm when consistent full-mark tool behavior outweighs the other axes.

Still has scripted safety warnings. Gyro-M low ties the observed floor but is unranked/experimental, not an equivalent production pick.

Fast coding loop · specialist override

S-IQ3_XXS (UkisAI) low · #9 overall

Complete coding wall 11m56s at 95.1%.

You are prioritizing latency, which is outside the composite; this is not a new all-round winner.

Highest stock coding ceiling · specialist override

Q-IQ3_S (ISTA-DASLab) xhigh · #7 overall

Complete reported coding 96.3% (158/164).

Scored blank/timeout failures remain. Gyro-S low ties this reported ceiling on rc1, but is not a tool-capable stock control.

Knowledge / debugging · specialist override

S-IQ3_S (UkisAI)

Hidden checks 20/25, algebra sample 85%. Low unranked (partial coding); xhigh #12 overall.

The partial low coding result prevents an overall crown. Specialist evidence is from separate debugging/knowledge samples; choose it only for that priority.

Experimental fork · no production recommendation

Gyro-M (AgentionAI) low · unranked rc2

HE+ 95.1%, observed floor 78.3, live tools. The board displays its results but does not authorize a stock rank.

Minimum 205 MiB free VRAM, 131,072 context, invalid-slice status. Not a supported 16 GB daily-driver pick. Gyro-S rc1 has no valid tools score.

FOR MAINTAINERS — measured findings and validity

- **Tool plumbing is build-specific:** rc1 forced calls were ignored with Gyro-S. rc2 live forced calls passed with Gyro-M on streaming and non-streaming. The historical finding stays scoped to rc1; Gyro-S on rc2 was not tested.
- **Experimental fork fit:** 74 resource samples, minimum 205 MiB free, peak 78 C. cuda_error=0, engine_restart=0, no_progress=0, stall_report=0, verify_timeout=0. This is not long-term stability proof.
- **Campaign status:** `completed_with_invalid_slices`, finished 2026-10-12T17:27:48+02:00. CTest 49 pass / 2 skip / 1 fail. ple_parity lacks its required Q2_0 PLE shard/capture fixtures; separate ple_reader_selftest passed.
- **Audit bookkeeping:** both HumanEval+ task audits ran; a dictionary merge overwrote intended audit-step names, making done.json mark them absent. Xhigh validity logic also missed blank finals. Raw status/scores are preserved; no silently upgraded clean pass.
- **Monitoring:** corrected busy-vs-down logic after the first aborted attempt. Perf-low/needle static-counter warnings were not accompanied by engine-event counters. Standalone effort probe is excluded from scores; possible timing interference is disclosed.
- **Operations:** production Swift IQ3_S restoration passed health/chat/tool/vision and re-enabled guards/updater. Report revisions did not alter services.

METHOD & CORRECTIONS — reproducible without hiding caveats

The PDF, GitHub README, scoreboard and recommendation registry are generated from the same extracted store. New data is integrated throughout, not isolated in a model-specific chapter. Source records and hashes are packaged; the portable compiler rebuilds from this folder.

- **Coding:** BenchKit direct, 164 attempted tasks, greedy temperature 0, presence penalty 1.0, one repair. Gyro-M timeout 900 seconds, no injected client token cap. Serving protocol and engine changes qualify historical cross-cohort comparisons.
- **Tools:** 2.7.1.dev16+gf7c34a130, 92 scenarios x five trials per effort; seed 42, temperature 0.0, timeout 300 s, turn cap 8. Separate Swift v2.8.0 / 97-scenario data is NOT mixed in.
- **Corrections:** historical partial coding denominators qualified; category scores changed to five-run means; inflated raw prefill removed; ceiling-minus-floor gap labeled correctly; peak temperature corrected from 64 C summary to measured 78 C. These corrections, not an unmeasured model promotion, change the earlier recommendation framing.
- **Hosted baseline protocol:** GPT-6-luna MAX is an OpenAI-hosted API reference, not a recommendation or a local run. MAX is explicitly pinned; unsupported/unverified Chat Completions sampling fields (temperature, presence_penalty, seed, top-p/top-k) are not forwarded by the Codex Responses bridge. The matched decode profile is invalid/incomplete and all hosted rates are withheld.
- **Not tested:** vision quality, multi-day agents, extra seeds, multi-GPU, other GPU sizes, full knowledge datasets, Gyro-M beyond 131K or Gyro-S on rc2. No automatic supported-memory or production-stability conclusion.

T11 · Version-pinned provenance

Provenance item	Pinned value
Stock engine digest	34CDE150B21148E6
Fork rc2 run	full-20261012-rerun-1230
Fork engine commit	`9ff02c9fa9c86a88088370925b3c9131cc5d6c26`
Fork binary SHA256	`08d70e30b9176dee5273ba74bd67c4d4c435bef92d449323168d7a8a3497585f`
Gyro-M model revision	`2cd4f24bcb0fa053149a4e4ec15506ecc64cafab`
Gyro-M model SHA256	`7aa36d15131d255179ab6c87fa9c9f81a0b541fc9e81e959d4c3e29eb20feb5f`
Fork context / KV	131,072 / int8
Rebuild	scripts/build_revision.py; data/ALL-METRICS.json; claims.json

T11 · 8 rows · end of table

PR #1850 · Strata · AgentionAI fork · Gyro source · Tool-eval-bench · EvalPlus · Five Bugs

Revised 2026-10-13 · source-qualified observations · all recommendations use the same registry and ranks. Numeric audit, whole-table pagination proof, source manifest and portable rebuild accompany the report.